RESEARCH FORUM PRESENTATION

THE EFFECTS OF REPEATED SPRINT TRAINING WITH VARYING INTER-REP REST INTERVALS (10s vs 20s) ON PERFORMANCE OUTCOMES IN SEMI-PROFESSIONAL MALE SOCCER ATHLETES

Presented by Palash Gupta

INTRODUCTION & NEEDS REVIEW

Players cover up to 9-16 km of Total Distance

High-Intensity Running and Sprinting accounting for up to 12% and 11%

83% of Goals are preceded by decisive actions such as Short Sprints or Jumping

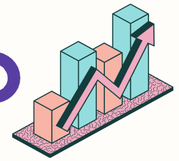
Demands
Repeated High-Intensity
Accelerations (30 ± 7, at > 3 ms^{-2})

Demands
Repeated High-Intensity
Decelerations (56 ± 12, at > 3 ms^{-2})

Demands
Changes of Direction
(~700 times per game)

Ability to perform Repeated High-Intensity Actions
Recover from Them
While Sustaining 90 min plus game

NEED TO DEVELOP



- Repeated Sprint Ability (RSA)
- Maximal Sprint Speed
- Acceleration & Deceleration
- Agility & Change of Direction
- Explosive Power
- Aerobic Capacity (VO2max)

HOW TO DEVELOP

Interval Training (for improvements in Aerobic Capacity, Fatigue Index, Lactate Threshold, and RSA Decrement)

Strength/Power Training (for improvements in Strength and Power)

High-Intensity Training (> VO2max, for improvements in Anaerobic Capacity)

Sprint Specific Training (for improvements in Sprint Speed and Acceleration)

PROBLEMS

Soccer demands players to manage congested match fixtures

Most of the time is dedicated to Technical and Tactical Training

Constraints on Pushing Players' Workload to elicit required adaptations

Less time available for prescribing multiple physical training modalities

PROPOSED SOLUTION:
RUNNING BASED REPEATED SPRINT TRAINING

KNOWN

Maximal effort, short-duration sprints (≤ 10 s) interspersed with brief (≤ 60 s) recovery periods
Standard Protocol: 3X6 at 30 m, 20 sec IRR, 4 min ISR

Saves Crucial Time & Develops both Neuromuscular and Metabolic Functions

Improves CMJ height, High-Intensity Running, Accels, Decels, Sprint Speeds, Sprint Times, Aerobic Capacity, Blood Lactate Buffer Capacity, CODs Ability, and RSA

Most studies compared (short vs long) W:R ratios (1:1 vs 1:6) and (1:3 vs 1:6)

UNKNOWN

Effects of narrower W:R ratios for Inter-Rep Rest Intervals compared to Standard Protocol such as (10s vs 20s | 1:2 vs 1:4)

Effects with Quasi Time & Sprint Volume Matched Protocols

Magnitude and Direction of Adaptions as per Rest Intervals

METHODS: SUBJECTS

A-Priori Power Analysis

- G*Power with α at 0.05, statistical power at 0.80, and an effect size of 0.65
- Total of 21 participants were required for this study
- 24 male Semi-Professional Soccer Players (EUAFC - Low Land League) will be invited to account for potential attrition and provide training stimulus to everyone

Inclusion Criteria

- Injury-free in lower body extremities in last 6 months
- Training history of a minimum of 3 years
- Consistently for at least 6 months before study entry

Additional Instructions

- Required to maintain a diet and training journal documenting the caloric intake along with the duration and perceived exertion of each day's workout
- 72 hours before testing, avoid alcohol, and in the 12 hours preceding testing, should not consume caffeine
- Subjects missing more than 20% of the training sessions will be excluded from further analysis.

METHODS: EXPERIMENTAL APPROACH

Study Design

- Randomised, Parallel Two Group (10s vs. 20s)
- Quasi-Time & Sprint-Volume Matched
- 6 Weeks Longitudinal RST Program (2 Sessions Per Week)
- Randomisation while accounting for Age, BMI, and Baseline Tests to minimise between-group baseline differences

Testing Schedule & Statistical Tests

- Pre-Post tests within 7 days of the intervention period under similar conditions
- Day 1: BMI, CMJ, Linear Sprint Speeds, and Sprint Times
- Day 2: MAS
- Tests included are valid and reliable as per the literature
- Paired T-Test & One-Way ANCOVA with Baseline Measurements as Covariates

METHODS: INTERVENTION PROTOCOL

# of Week	Sets	Reps	Sprint Distance	Inter-Rep Rest Interval Group 1 (10-sec)	Inter-Set Rest Interval Group 1 (80-sec)	Inter-Rep Rest Interval Group 2 (20-sec)	Inter-Set Rest Interval Group 2 (40 - 54 sec)	Total Time For Both Groups	Total Sprint Volume 2 x Week
1	4	3	35-m	10-sec	80-sec	20-sec	53-54 sec	320-sec	840-m
2	4	3	35-m	10-sec	80-sec	20-sec	53-54 sec	320-sec	840-m
3	4	3	35-m	10-sec	80-sec	20-sec	53-54 sec	320-sec	840-m
4	4	4	35-m	10-sec	80-sec	20-sec	40-sec	360-sec	1120-m
5	4	4	35-m	10-sec	80-sec	20-sec	40-sec	360-sec	1120-m
6	4	4	35-m	10-sec	80-sec	20-sec	40-sec	360-sec	1120-m

- Sessions will be scheduled for Monday & Thursday and will replace regular physical conditioning sessions
- Participants will be supervised through a 10-minute dynamic warm-up using the RAMP protocol
- Perform 2 progressive sprints of 35 m at 70% and 90% of perceived maximal effort
- Sessions will occur on artificial turf, and instructors will provide verbal encouragement

METHODS: PERFORMANCE TESTS

Linear Sprint Speeds & Times (20 m & 40m)

- Mean and Peak Velocities will be assessed using Electronic Timing Gates (10-m Split) and 10-Hz GPS on artificial turf
- Stand in a 2-point position on a point 0.5 m behind the timing gate with a 10-second warning and 3-second countdown
- Best of 3 trials with 3 min between trial passive recovery

Maximal Aerobic Speed

- Measured through 1600 m time trial using a stopwatch
- MAS = TTavgspeed * (0.766 + (0.117 * TTdistance))

Countermovement Jump Height

- Measured using Dual Force Plates
- Best of 3 trials with 30 seconds recovery. 4th Trial if CMJH shows uptrend
- Calculated using the Impulse Momentum approach (TTTO and CM Depth)

Internal Training Load

- Measured using HR and dRPE (sRPE-B & sRPE-LE) using CR100 Scale
- dRPE will be collected at the middle & end of the last repetition of the protocol

ADJUSTMENTS AS PER RESEARCH PROCESS FEEDBACK

Comment on the construct of Repeated Sprint Ability as DV	Reviewed the literature and changed the narrative of RST/A as IV instead of DV and dropped RSA Test
Comment on emphasising the time efficiency in soccer	Feedback implemented highlighting the time-efficient nature of RST
Initial Proposal 15s vs. 20s | 1:3 vs. 1:4	Changed to 10s vs. 20s | 1:2 vs. 1:4 For Practical and Logistical Reasons
Comment on the narrative of second research question	Dropped the second research question as it was redundant

ADJUSTMENTS AS PER RESEARCH PROCESS FEEDBACK

Initial Proposal
Blinding Participants

→ Blinding Not Required

Comment on the use of ANOVA and between group baseline comparison

→ Changed to ANCOVA & dropped baseline between group comparisons

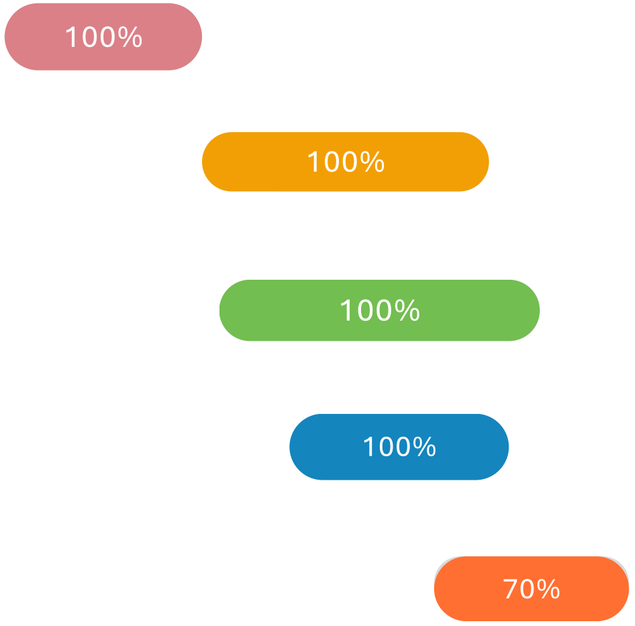
Initial Proposal
Conduct Reliability using ICC and SEM

→ Will be using Literature Derived Values

Comment on monitoring training load outside sessions

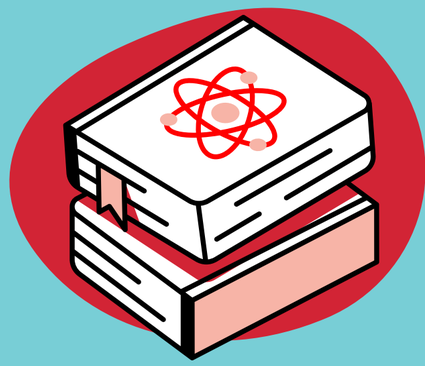
→ Participants will be instructed to maintain training journal with session RPE and duration

PROJECT UPDATES & EQUIPMENT REQUIREMENTS



Measurement Metrics	Equipment
Height	Stadiometer
Body Mass	Scales
Training and Testing Setup	Measurement Tape, Cones, Access to Track or Artificial Turf
Heart Rate	Heart Rate Monitor
Sprint Times	Electronic Timing Gates
dRPE	CR100 Scale
Maximal Aerobic Speed	Stopwatch
Linear Sprint Speed	Electronic Timing Gates and GPS
Countermovement Jump Height	Dual Force Plates
Environmental Conditions	Temperature, Wind, and Humidity using Weather Monitoring Device
Subjects	Minimum of 21

THANK YOU



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